A small checked argument

Your Name

Abstract. We prove that every proposition implies itself. A Lean proof and its intermediate goals show how the argument works, one step at a time.

1. A definition

We begin with a small definition. The browser export lets the reader inspect its type and documentation directly from the highlighted name.

```
def hello : String := "Hello from Lean!"
```

2. A checked proof

For arbitrary propositions P and Q, each proposition implies itself. We prove both implications together by constructing a conjunction.

```
theorem identity_pair (p q : Prop) :
    (p → p) ∧ (q → q) := by
  constructor
  all_goals
    intro h
    exact h
```

The tactic `constructor` creates two goals. The `all_goals` block introduces the hypothesis in each branch and then uses it to finish the proof.

3. Inspecting the goals

The state after `constructor` makes the two branches explicit:

```
case left
p : Prop
q : Prop
⊢ p → p

case right
p : Prop
q : Prop
⊢ q → q
```

Proof-state bubbles in the quotation provide the remaining intermediate states.

4. Checking the result

Lean reports the theorem's type and evaluates a small expression:

```
#check Demo.identity_pair
```

```
Demo.identity_pair (p q : Prop) : (p → p) ∧ (q → q)
```

```
#eval (2 + 2)
```

```
4
```